

## TASK 2: SALES PERFORMANCE ANALYSIS

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**GITHUB LINK:** <https://github.com/Giridharan2007/Sales-Analysis>

### Objective

The objective of this project is to analyze sales performance across different regions, product categories, and time periods using the Superstore Sales dataset. The analysis aims to identify business trends, top-performing products, low-performing products, and seasonal sales patterns.

### Dataset

Source: Kaggle Superstore Sales Dataset

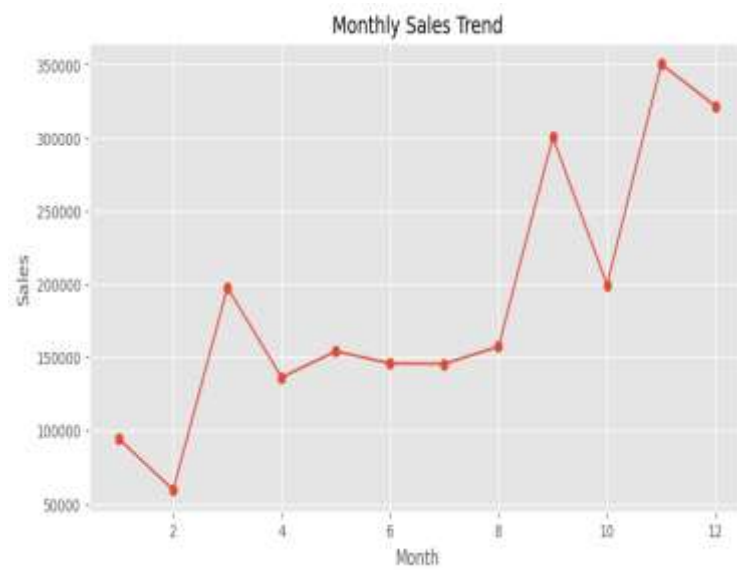
The dataset contains information related to orders, sales, profit, customers, categories, sub-categories, regions, discounts, and shipping details.

### Business Recommendations

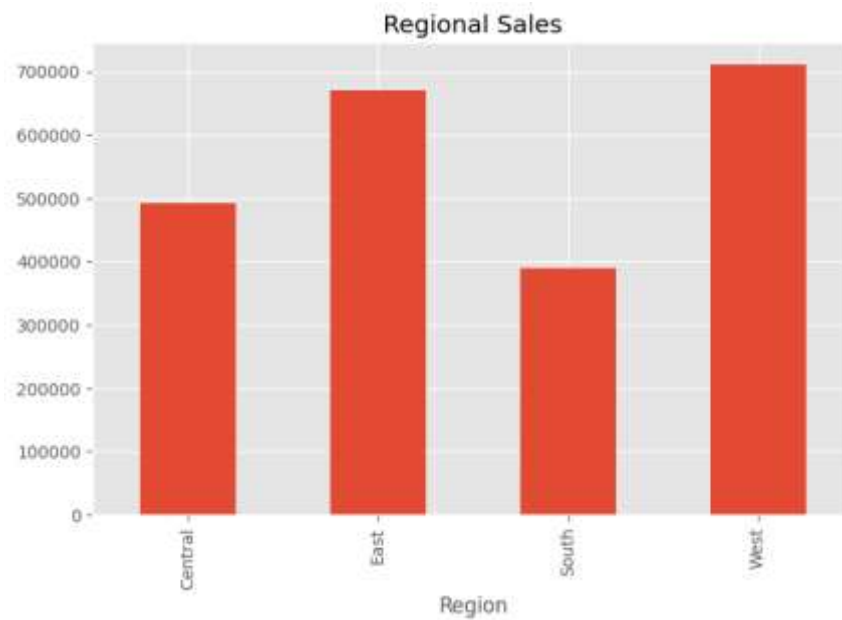
1. Increase promotion of high-profit Technology products.
2. Optimize inventory for low-performing products.
3. Strengthen marketing in low-performing regions.
4. Launch seasonal campaigns before peak sales months.
5. Implement cross-selling and upselling strategies.

## Visualizations Included

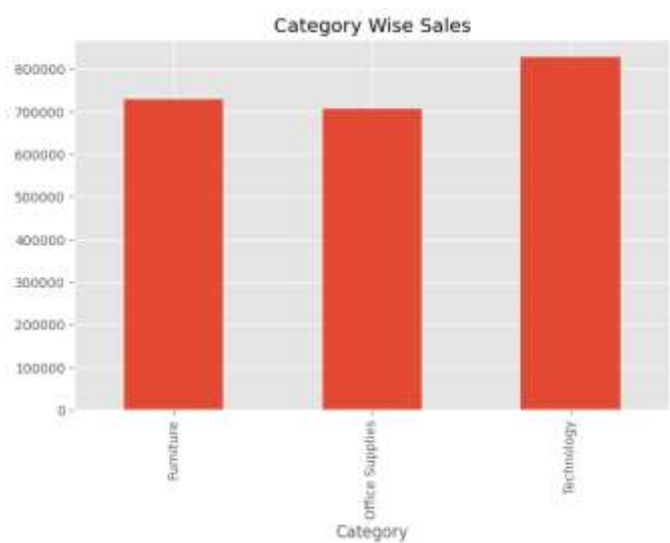
- Monthly Sales Trend



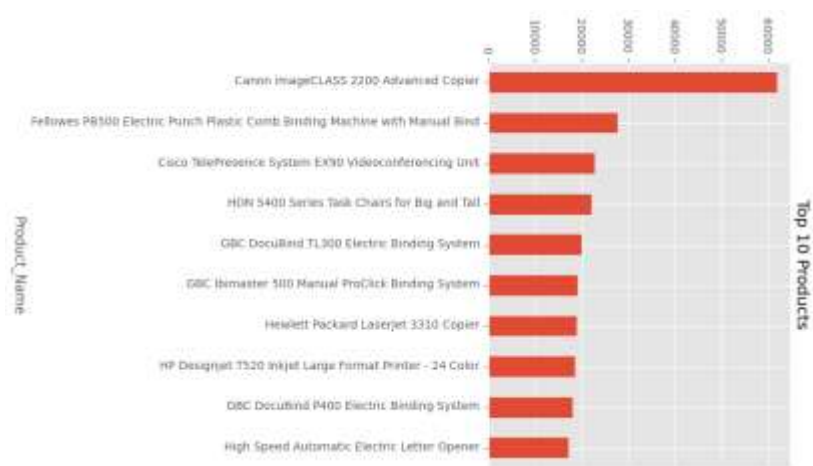
- Regional Sales Comparison



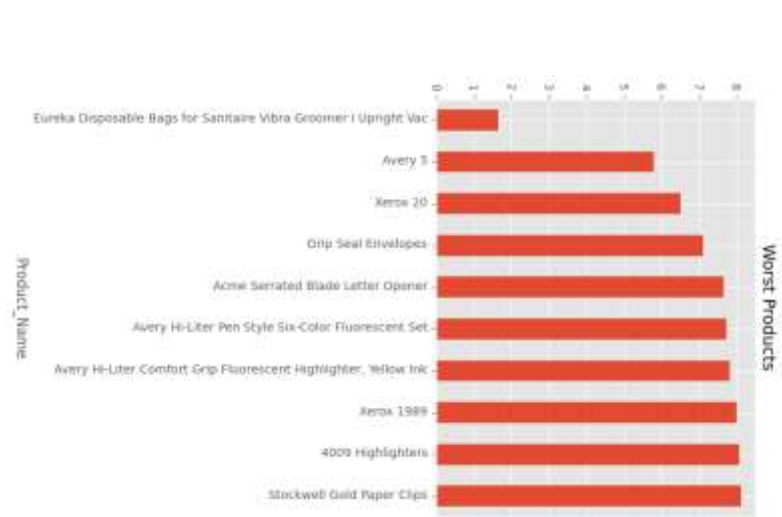
- Category-wise Sales



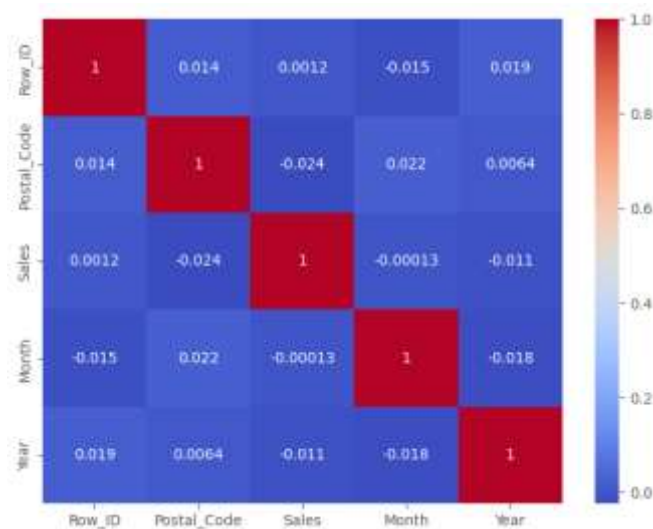
- Top 10 Products



- Bottom 10 Products



- Correlation Heatmap



## Tools Used

1. **Python** – Data analysis and processing
2. **Google Colab** – Running Python code online
3. **Pandas** – Data cleaning and manipulation
4. **NumPy** – Numerical calculations
5. **Matplotlib** – Data visualization and charts
6. **Seaborn** – Statistical data visualization
7. **Scikit-learn** – Customer segmentation using K-Means clustering
8. **Jupyter Notebook / Google Colab** – Interactive coding environment
9. **Microsoft Excel** – Initial dataset review and validation
10. **Kaggle Dataset** – Data source for analysis

## Conclusion

The analysis successfully identified sales trends, profitable business segments, and seasonal patterns. The findings provide actionable insights that can help improve revenue, profitability, and customer engagement. Strategic implementation of the recommendations can support sustainable business growth.